Name:		Date:		Period:	

Goldfish Paradise: Ecosystem Lab Worksheet

In this lab you will use a computer model of a fish tank to find out what keeps an aquatic ecosystem healthy, why ecosystems sometimes collapse, and whether growing plants on fish water can clean it up and grow food.

Getting started

1. Open the lab link from your teacher. Make sure Basic is selected (top right).
2. Click the Missions tab. Each mission has a button that sets up the tank for you.
3. Use Run, then read the status cards, the What happened? list and the charts.
4. The Look at day slider lets you see the tank on any day.

Remember: this is a model. It follows real science, but its numbers are not measurements from a real tank.

Words to know

Word	What it means
Producer	A living thing that makes its own food from sunlight (photosynthesis). Algae and plants.
Consumer	A living thing that eats other things for energy. The goldfish.
Decomposer	Bacteria that break down dead things and waste. They use oxygen while they work.
Photosynthesis	Producers use light, CO_2 and water to make food and give off oxygen. Only in light.
Respiration	All living things use food and oxygen for energy and give off CO_2. Day AND night.
Ammonia	Toxic waste from fish. Bacteria change it into nitrate (nitrification).
Nitrate	A less harmful form of nitrogen. Algae and plants use it to grow.
Algal bloom	A burst of algae growth that turns the water green.
Limiting factor	The thing in shortest supply that stops a population from growing more.

Mission 1: Keep the fish alive

Eight goldfish live in this tank. In the Missions tab, press Load the starting tank. The settings are: 8 fish, Normal food, Medium light, Air pump Off, 60 days.

Prediction (before you press Run)

After 60 days I think: ☐ all 8 fish live ☐ some fish die ☐ all the fish die

Because...

Data table

Trial 1 uses the starting settings. For each new trial, change only ONE thing from Trial 1.

Trial	What I changed	Fish alive at day 60	Lowest oxygen (G / Y / R)	Water color
1	Nothing (starting tank)			
2				
3				
4				
5				
6				

G / Y / R = the color of the Oxygen status card: green (Healthy), yellow (Getting low), red (Fish in danger). Use the Look at day slider to find the worst day. Water color: clear, green, or pea soup.

Questions

1. Which change(s) kept all 8 fish alive? Did more than one change work?

2. Pick one change that worked. Explain why it helped. Use the word oxygen.

Sentence starter: When I changed _____, the fish survived because...

Mission 2: Why did the fish die?

Load the starting tank again and press Run. Read the What happened? list.

A. When did it happen?

Event	Day
The water turned green (algae growing fast)	
Decomposer bacteria multiplying fast	
Oxygen dropped below the danger line at night	
The first fish died	
Half of the fish were gone	

B. Put the chain in order

Number the boxes 1 to 6 to show what caused what. Box 1 starts the chain.

___ Decomposers multiply and use up oxygen	___ Fish die	___ Food and fish waste add nutrients to the water
___ Oxygen drops very low at night	___ Algae grow fast (algal bloom)	___ Algae die and waste piles up

C. Questions

1. Look at the oxygen chart. Is oxygen lowest in the day or at night? Why?

Hint: which process happens only in the light, and which happens all the time?

2. Algae make oxygen. So how can a huge algal bloom end with too little oxygen?

3. Did the fish die mostly from low oxygen or from toxic ammonia? (Look at the last line of What happened?)

Mission 3: Waste to Plate

In aquaponics, fish water flows past plant roots. The plants soak up nitrogen from fish waste and grow food. In the Missions tab, press Load the comparison. Both tanks get the same fish, food, light and air pump.

Prediction

I think the tank with plants will have: ☐ more waste ☐ less waste ☐ the same waste

I think the plants will: ☐ get rid of ALL the algae ☐ reduce the algae ☐ not change the algae

Data table

Read the values at day 60. Waste = the "fish waste" chart. Food grown = the status card on the plants tank.

Run	Settings	Tank	Fish alive	Waste at day 60 (mg/L)	Water	Food grown (g)
1	Normal food, pump ON	Fish tank				—
		+ plants				
2	A lot of food, pump ON	Fish tank				—
		+ plants				
3	A lot of food, pump OFF	Fish tank				—
		+ plants				

Questions

1. In Run 1, how much less waste did the plants tank have? Did the plants get rid of all the algae?

2. In Run 3 the air pump was off. Did the plants still save the fish? Why not?

Hint: plant leaves are above the water. Where does the oxygen they make go?

Claim, Evidence, Reasoning

Question: Can aquaponics (Waste to Plate) reduce the impact of fish waste on a water ecosystem?

Claim

Sentence starter: Aquaponics can / cannot reduce the impact of fish waste because…

Evidence

Sentence starter: In Run ___, the tank with plants had ___ mg/L of waste and the fish tank had ___ mg/L. Also…

Reasoning

Sentence starter: This happens because plants use nitrogen from the water to… When we harvest the plants, the nitrogen…

Matter and energy check

Statement	True	False
When bacteria turn ammonia into nitrate, the nitrogen leaves the tank.	☐	☐
The carbon in a fish can later become part of an algae cell.	☐	☐
Energy is recycled in an ecosystem, just like carbon and nitrogen.	☐	☐
When we harvest the plants, nitrogen leaves the system as food.	☐	☐
Fish use most of the energy in their food to stay alive, and it ends up as heat.	☐	☐

Reflection

Give one way the computer model is different from a real fish tank or a real river like the Saginaw River.